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Connector having a hood for protecting a cable pull-out portion

Abstract

A module is composed of a terminal that is attached to a cable and an insulator that accommodates the terminal. The insulator includes two spring pieces and two protrusions that are formed on the two respective spring pieces. A housing has through holes on mutually opposed positions on respective side walls surrounding an accommodation space. The module is accommodated in the accommodation space in a manner such that the two protrusions thereof are fitted from an inside into the respective through holes. A hood is composed of two split bodies. The hood can be fixed on the housing in a manner such that projecting portions of the split bodies are fitted from an outside into the respective through holes and the two split bodies are coupled to each other with a coupling means.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a connector that includes a hood for protecting a cable pull-out portion.
BACKGROUND ART
(2) FIGS. **1** and **2** illustrate a configuration described in Japanese Patent Application Laid Open No. 2016-195011 as a related art example of the configuration in which a hood (connector hood) is attached to a connector. In FIG. **1**, **10** denotes a hood and **13** denotes a cable. Further, **20** denotes a mass flow controller (MFC) to which the connector is connected.
(3) FIG. **2** illustrates an internal structure of the hood **10** of FIG. **1**. The hood **10** is connected to a connector terminal **11** provided on an upper surface of the MFC **20**. The hood **10** includes a first member **1**, a second member **2**, a third member **3** (see FIG. **1**, illustration is omitted in FIG. **2**), a

fourth member **4** including an opening for the connector terminal **11**, side surface screw **5**, and upper surface screw **6**. The side surface screw **5** serve to couple the first to fourth members **1** to **4** and the like to each other, and the upper surface screw **6** serve to fix the connector terminal **11** held by the hood **10** to the MFC **20**.
(4) The connector terminal **11**, wires **12**, the cable **13**, and a bushing **14** are contained in the inside of the hood **10**, and the wires **12** are connected to respective pins of the connector terminal **11**.
(5) The above-mentioned configuration of the related art uses screws for attaching and fixing a hood and accordingly requires forming the screw holes for the screws in a fixing target (the MFC **20** in the above-mentioned example) of the hood. That is, related art configurations have required an additional work for forming special screw holes in the fixing target for attaching and fixing the hood.
BRIEF SUMMARY OF THE INVENTION
(6) An object of the present invention is to provide a connector that includes a hood which can be attached without a special means that is additionally required for attaching the hood.
(7) According to the present invention, a connector includes a housing, a module, and a hood. The module is composed of a terminal, which is attached to an end of a cable and connected with a mating terminal of a mating connector, and an insulator which accommodates and holds the terminal. The insulator includes two spring pieces, which are formed on respective outer surfaces that are positioned on mutually opposite sides, and two protrusions, which are formed on the two spring pieces respectively in a manner to protrude mutually outward. The housing has an accommodation space and a through hole is formed on each of mutually opposed positions on respective side walls surrounding the accommodation space. The module is positioned and accommodated in the accommodation space to be attached to the housing in a manner such that each of the two protrusions is fitted from an inside into the through hole, which is formed on each of the mutually opposed positions on the side walls. The hood is composed of two split bodies that can be separated from and joined to each other, and a projecting portion is formed on each of the two split bodies. The hood can be positioned and fixed on the housing in a manner such that the projecting portion of each of the two split bodies is fitted from an outside into the through hole, which is formed on each of the mutually opposed positions on the side walls, and the two split bodies are coupled to each other with a coupling means. When a connecting direction between the terminal and the mating terminal is defined as an up and down direction, the two split bodies have respective shapes to form a straight pull-out hole through which the cable can be pulled out from the hood upward and form an angle pull-out hole through which the cable can be pulled out from the hood in a direction orthogonal to the up and down direction, in a state in which the hood is positioned and fixed on the housing.
Effects of the Invention
(8) According to the present invention, it is possible to obtain a connector that includes a hood which can be easily attached without newly requiring a special means for attaching the hood.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating a related art configuration in which a hood is attached to a connector.
(2) FIG. **2** is a perspective view illustrating chief-component internal structure of FIG. **1**.
(3) FIG. **3**A is a front perspective view illustrating an example of a connector according to the present invention.
(4) FIG. **3**B is a rear perspective view illustrating the example of the connector according to the present invention.

(5) FIG. **4** is a longitudinal sectional view of the connector illustrated in FIG. **3**A.
(6) FIG. **5** is a perspective view illustrating a state in which a hood is removed from the connector illustrated in FIG. **3**A.
(7) FIG. **6** is a longitudinal sectional view of the state illustrated in FIG.
(8) FIG. **7**A is an upper perspective view illustrating one split body of a hood.
(9) FIG. **7**B is an upper perspective view illustrating the other split body of the hood.
(10) FIG. **7**C is a lower perspective view illustrating one split body of the hood.
(11) FIG. **7**D is a lower perspective view illustrating the other split body of the hood.

LIST OF REFERENCE NUMERALS

(12) **1**: first member **2**: second member **3**: third member **4**: fourth member **5**: side surface screw **6**: upper surface screw **10**: hood **11**: connector terminal **12**: wire **13**: cable **14**: bushing **20**: MFC **30**: housing **31**: accommodation space **32**: side wall **33**: side wall **34**: side wall **35**: side wall **36**: through hole **37**: lever **40**: module **41**: terminal **41***a*: cut-and-raised piece **42**: insulator **43**: attaching hole **43***a*: step portion **44**: spring piece **44***a*: protruding portion **44***b*: protrusion **50**: hood **51**: straight lead-out hole **60**: split body **61**: side surface plate portion **62**: upper surface plate portion **63**: front surface plate portion **63***a*: inclined portion **64**: back surface plate portion **65**: cutout **66**: cutout **67**: projecting portion **68**: bundling-band threading hole **69**: fixing portion **70**: split body **71**: side surface plate portion **72**: upper surface plate portion **73**: front surface plate portion **73***a*: inclined portion **74**: back surface plate portion **75**: cutout **76**: cutout **77**: projecting portion **78**: bundling-band threading hole **79**: fixing portion **80**: bundling band **90**: grommet **100**: connector **200**: cable **201**: core wire

DETAILED DESCRIPTION

(13) An embodiment of the present invention will be described based on an example with reference to the accompanying drawings.
(14) FIGS. **3**A and **3**B illustrate an outer appearance of an example of a connector according to the present invention, and FIG. **4** illustrates a cross-sectional structure of the connector. A connector **100** in this example is configured to include a housing **30**, modules **40**, and a hood **50**. Here, in FIGS. **3**A, **3**B, and **4**, **200** denotes cables connected to the connector **100**, and the cables **200** are only partially illustrated in this example, with the entire illustration thereof omitted.
(15) FIG. **5** illustrates a state in which the hood **50** is removed from the connector **100**, and FIG. **6** illustrates a cross-sectional structure of the connector **100** in this state. Configurations of the housing **30** and the module **40** will be first described with reference to FIGS. **5** and **6**.
(16) The housing **30** has a substantially quadrangular frame shape and has an accommodation space **31** thereinside that is open vertically. The accommodation space **31** is surrounded by four side walls **32** to **35**. The side walls **32** and **33**, which are in the longer sides of the quadrangular frame shape, are formed with through holes **36** in their upper end portions. The through hole **36** has an angled hole shape and communicates the inside and outside of the housing **30**. In this example, six through holes **36**, respectively, are aligned in an extending direction of the longer sides on each of the side walls **32** and **33**. The through holes **36** are formed in mutually opposed positions on the side walls **32** and **33**. Although detailed description is omitted, the housing **30** includes a lever **37** for locking to be used in mating with a mating connector.
(17) The module **40** is composed of terminals **41** and an insulator **42**. The terminals **41** are attached to ends of the cables **200** and can be connected with mating terminals of a mating connector. The insulator **42** accommodates and holds the terminals **41**. In this example, the insulator **42** has four attaching holes **43** that penetrate vertically, and four terminals **41** are attached in these attaching holes **43**. The terminal **41** is configured to be inserted into the attaching hole **43** from the upper side and retained as a cut-and-raised piece **41***a* of the terminal **41** is caught on a step portion **43***a* formed in the attaching hole **43**. Here, a core wire **201** of the cable **200** is crimped and connected to the upper end of the terminal **41**.
(18) The outer surfaces of two mutually opposite sides of the insulator **42** are formed with spring

pieces **44** that extend, respectively, along the outer surfaces. A lower end of the spring piece **44** is a fixed end and an upper end is a free end. Protruding portions **44***a*, which protrude mutually outward, are formed on the respective free ends of the two spring pieces **44**, and protrusions **44***b*, which protrude mutually outward, are further formed in the vicinity of the respective free ends.

(19) The module **40** having the above-described configuration is attached in the accommodation space **31** of the housing **30**. The attachment is performed by inserting the module **40** into the accommodation space **31** from the upper side of the accommodation space **31**. The module **40** is positioned and accommodated in the accommodation space **31** to be attached to the housing **30** in a manner such that the two protrusions **44***b*, which are formed on the respective two spring pieces **44**, are fitted from the inside to the respective through holes **36**, which are formed on the mutually opposed positions on the side walls **32** and **33**. Here, a pair of the protruding portion **44***a* and the protrusion **44***b* that are formed on the spring piece **44** pinches a portion of the side wall **32** or **33** that is higher than the though hole **36** (i.e., its upper end portion), as illustrated in FIG. **6**.

(20) The module **40** is attached to the housing **30** as described above. In this example, six pairs of through holes **36**, each pair of which are mutually opposed, are formed in the side walls **32** and **33** of the housing **30** and therefore, up to six pieces of modules **40** can be positioned and accommodated in the accommodation space **31** of the housing **30**. Here, the drawings illustrate the state in which six pieces of modules **40** with identical specifications are accommodated, in this example. However, the number of modules **40** to be accommodated, specifications of the terminals **41** provided to the module **40**, and further, specifications and the number of attaching holes **43** of the insulator **42** to which the terminals **41** are to be attached are appropriately determined depending on a specification and the number of cables **200** connected by the connector **100**.

(21) The hood **50** included in the connector **100** will now be described.

(22) The hood **50** is composed of two split bodies **60** and **70** that can be separated from and joined to each other. FIGS. **7**A and **7**B illustrate these split bodies **60** and **70** respectively, and FIGS. **7**C and **7**D illustrate the split bodies **60** and **70** respectively viewed in a different direction.

(23) The split bodies **60** and **70** have mutually plane-symmetrical shapes in this example; the split body **60** constitutes a left half portion of the hood **50**, and the split body **70** constitutes a right half portion of the hood **50**.

(24) The split body **60** is, mainly, composed of a side surface plate portion **61**, an upper surface plate portion **62**, a front surface plate portion **63**, and a back surface plate portion **64**. The split body **70** is similarly composed of a side surface plate portion **71**, an upper surface plate portion **72**, a front surface plate portion **73**, and a back surface plate portion **74**. On the upper surface plate portions **62** and **72** of the split bodies **60** and **70**, semicircular cutouts **65** and **75** are formed respectively. The cutouts **65** and **75** form a circular hole when the split bodies **60** and **70** are abutted on each other with the side surface plate portions **61** and **71** positioned outside. In a similar manner, semicircular cutouts **66** and **76**, which are to form a circular hole, are formed on inclined portions **63***a* and **73***a* of the front surface plate portions **63** and **73** respectively.

(25) On respective lower ends of inner surfaces of the side surface plate portions **61** and **71** in these split bodies **60** and **70**, projecting portions **67** and projecting portions **77** are formed so as to project mutually inward. In this example, three projecting portions **67** are formed closer to the front surface plate portion **63** and one projecting portion **67** is formed closer to the back surface plate portion **64**. In a similar manner, three projecting portions **77** are formed closer to the front surface plate portion **73** and one projecting portion **77** is formed closer to the back surface plate portion **74**. These projecting portions **67** and **77** can be fitted to the through holes **36**, which are formed so as to be aligned and positioned on mutually opposed positions on the side walls **33** and **32** of the housing **30**.

(26) Further, bundling-band threading holes **68** and **78** are formed on the split bodies **60** and **70** respectively. The bundling-band threading holes **68** and **78** are used for fastening the split bodies **60** and **70** to each other with bundling bands. The bundling-band threading holes **68** are formed in

respective four fixing portions **69** and the bundling-band threading holes **78** are formed in respective four fixing portions **79**, as illustrated in FIGS. **7**A, **7**B, **7**C, and **7**D. The fixing portions **69** and the fixing portions **79** are formed on outer surfaces of the split bodies **60** and **70** in a protruded manner.
(27) Attachment of the hood **50** will be described below.
(28) The four projecting portions **67** of the split body **60** are fitted from the outside into the respective through holes **36**, which are formed so as to be aligned on the side wall **33** of the housing **30**, and, in the same manner, the four projecting portions **77** of the split body **70** are fitted from the outside into the respective through holes **36**, which are formed so as to be aligned on the side wall **32** of the housing **30**. The split bodies **60** and **70** are thus attached to the housing **30**.
(29) Subsequently, a bundling band **80** is threaded through each one of the bundling-band threading holes **68** of the split body **60** and corresponding one of the bundling-band threading holes **78** of the split body **70** so as to fasten the split bodies **60** and **70** to each other with the bundling bands **80**. Since the split body **60** and **70** are formed with the bundling-band threading holes **68** and **78** in **4** places, respectively, they are fastened to each other with four bundling bands **80**. The hood **50** is thus positioned and fixed on the housing **30**, being attached to the housing **30**. That is, the hood **50** comes to be in the state illustrated in FIGS. **3**A, **3**B, and **4**.
(30) In the state in which the split bodies **60** and **70** are fastened to each other and the attachment of the hood **50** to the housing **30** is completed as illustrated in FIGS. **3**A and **3**B, a straight lead-out hole **51** having a circular shape is formed by the cutout **65** of the split body **60** and the cutout **75** of the split body **70**, and an angled lead-out hole having a circular shape is formed by the cutout **66** of the split body **60** and the cutout **76** of the split body **70**. These straight lead-out hole **51** and angled lead-out hole have mutually-equal shapes and dimensions in this example.
(31) The connector **100** is configured to be mated with a mating connector that is positioned underneath the housing **30**. When a connecting direction between the terminals **41** and the mating terminals of the mating connector is defined as an up and down direction, the cables **200** can be led out of the hood **50** upward using the straight lead-out hole **51** formed on the hood **50**, or in a direction orthogonal to the up and down direction using the angled lead-out hole. FIGS. **3**A and **3**B illustrate the state in which the cables **200** are led out of the hood **50** upward, and in this example, a grommet **90** is attached on the angled lead-out hole and the angled lead-out hole is closed by the grommet **90**.
(32) The example of the connector according to the present invention has been described above, and according to the connector **100** described above, following advantageous effects can be obtained.
(33) (1) The hood **50** is attached to the housing **30** by utilizing the through holes **36** formed on the side walls **32** and **33**, which are configured to be used to position and attach the modules **40**. Specifically, the hood **50** is attached by using outer side portions of the through holes **36** that are not occupied by the protrusions **44***b* of the modules **40**. Thus, there is no need to additively provide a special means for attaching the hood **50**, and the hood **50** can be easily attached.
(34) (2) The hood **50** is composed of two split bodies **60** and **70**. The split bodies **60** and **70** are attached to the housing **30** and then fixed with the bundling bands **80** to complete the attachment of the hood **50**. Thus, the attaching work can be simplified by using the bundling bands **80**.
(35) (3) The hood **50** has two holes that are the straight lead-out hole **51** and the angled lead-out hole as holes to lead out the cables **200** through. Therefore, a direction in which the cables **200** are led out can be selected depending on a using environment and a using condition of the connector **100**, and further, if circumstances require, the cables **200** can be led out in both of the two directions by using both of the two holes. In that way, the connector **100** serves as a useful connector.
(36) In the above-described example, the two split bodies **60** and **70**, which constitute the hood **50**, are formed with the four projecting portions **67** and **77** to be fitted into the through holes **36** of the

housing **30**, respectively. However, the numbers of projecting portions **67** and **77** are not limited to the above and each of the split bodies **60** and **70** may have, for example, a single projecting portion.
(37) Further, in the above-described example, the angled lead-out hole that is not used for leading out the cables **200**, is closed with the grommet **90**. However, an embodiment in which the angled lead-out hole is not closed with the grommet **90** may be employed depending on a situation.
(38) Furthermore, the bundling bands **80** are used there as the fastening means for fastening together the two split bodies **60** and **70** of the hood **50**. However, the fastening means is not limited to this and the two split bodies **60** and **70** may be, for example, screwed together with screws.
(39) Here, the connector according to the present invention includes the hood **50** having the above-described configuration, as a component. However, when the hood **50** is not necessary, a form in which the hood **50** is not attached may be employed.
(40) The foregoing description of the embodiment of the invention has been presented for the purpose of illustration and description. It is not intended to be exhaustive and to limit the invention to the precise form disclosed. Modifications or variations are possible in light of the above teaching. The embodiment was chosen and described to provide the best illustration of the principles of the invention and its practical application, and to enable one of ordinary skill in the art to utilize the invention in various embodiments and with various modifications as are suited to the particular use contemplated. All such modifications and variations are within the scope of the invention as determined by the appended claims when interpreted in accordance with the breadth to which they are fairly, legally, and equitably entitled.

Claims

1. A connector to be attached to an end of a cable and used for mating with a mating connector having a mating terminal, the connector comprising: a housing having an accommodation space therein, the housing comprising side walls surrounding the accommodation space, the side walls having a pair of through holes formed at mutually opposed positions thereof, the housing having an inside and an outside thereof associated with the accommodation space; a module comprising a terminal and an insulator that holds and houses the terminal, the terminal being configured to be attached to the end of the cable and capable of connecting the mating terminal, the insulator having two outer surfaces positioned at mutually opposite sides thereof, two spring pieces being provided on the two outer surfaces respectively, two protrusions being formed on the two spring pieces respectively, the two protrusions protruding mutually outward, wherein the module is attached to the housing such that the module housed in the accommodation space is positioned by fitting of the two protrusions into the pair of through holes from the inside of the housing; and a hood comprising two split bodies that are configured to be fastened to each other by a fastening means, two projecting portions being formed on the two split bodies respectively, wherein the hood is capable of being fixed to the housing in a manner such that the hood is positioned on the housing by fitting of the two projecting portions into the pair of through holes from the outside of the housing while the two split bodies are fastened to each other with the fastening means, and wherein given that a connecting direction between the terminal and the mating terminal is defined as an up and down direction, the two split bodies have shapes such that a straight lead-out hole through which the cable can be led out of the hood upward and an angled lead-out hole through which the cable can be led out of the hood in a direction orthogonal to the up and down direction, are formed in a state in which the hood is positioned and fixed on the housing.
2. The connector according to claim 1, wherein the side walls have another one or more pairs of through holes adapted for attaching another one or more modules to the housing, each of said another one or more through holes being formed at another two mutually opposed positions of the side walls.
3. The connector according to claim 2, wherein the straight lead-out hole and the angled lead-out

hole have mutually-equal shapes and dimensions, and the connector further comprises a grommet that can close arbitrary one of the straight lead-out hole and the angled lead-out hole.

4. The connector according to claim 2, wherein each of the two split bodies is formed with a bundling-band threading hole, and the two split bodies can be fastened to each other by a bundling-band as the fastening means that is threaded through the bundling-band threading holes of the two split bodies.

5. The connector according to claim 1, wherein the straight lead-out hole and the angled lead-out hole have mutually-equal shapes and dimensions, and the connector further comprises a grommet that can close arbitrary one of the straight lead-out hole and the angled lead-out hole.

6. The connector according to claim 1, wherein each of the two split bodies is formed with a bundling-band threading hole, and the two split bodies can be fastened to each other by a bundling-band as the fastening means that is threaded through the bundling-band threading holes of the two split bodies.